

v0.3.8 · SUBMISSION BASELINE

PRD Genie

An evidence-governed multi-agent pipeline
from source documents to planning-ready delivery.

Vipin Puri · IK S2 Capstone Program · August 2026

Source evidence → Requirements → PRD → Stories → Validation → Sizing → Delivery

Governed evidence lineage

A governed pipeline turns fragmented evidence into traceable delivery

THE CHALLENGE

Manual reconciliation slows delivery

Transcripts, briefs, notes, and stakeholder decisions are reconciled by hand—creating delay, inconsistency, and weak source traceability.

THE GOVERNED RESPONSE

Specialized agents preserve evidence

Nine coordinated workflows ingest evidence, extract requirements, expose gaps, obtain human approval, generate artifacts, validate lineage, and size stories.

THE VALIDATED OUTCOME

Planning-ready artifacts with controls

- Corrected ten-section PRD
- 3 Epics · 7 Features · 11 Stories
- Acceptance criteria for all stories
- 11/11 stories sized
- Zero orphaned governed records

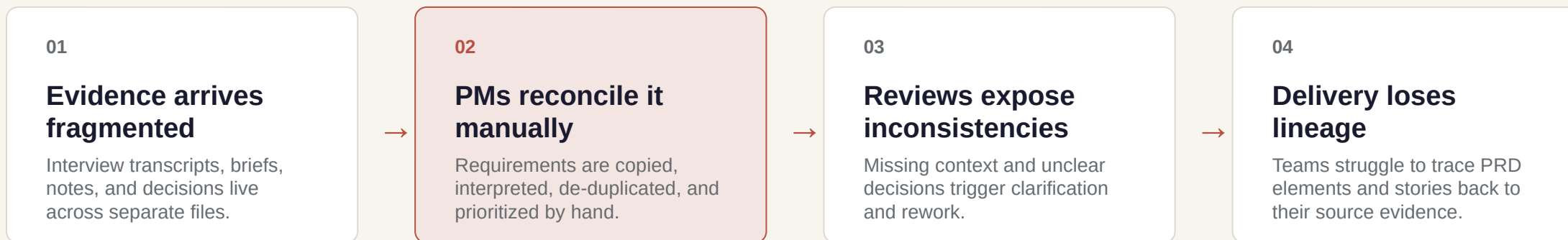
v0.3.8 accepted submission candidate · Agreement Gate release authorized

EVIDENCE AT A GLANCE

6 authoritative sources → 9 coordinated workflows → 3 / 7 / 11 hierarchy → 7 validated artifacts

Manual documentation creates delay and traceability risk

At NeuronForge, product managers must translate fragmented stakeholder evidence into decisions that engineering can trust.



WHO FEELS THE FRICTION

Product managers · Engineering leads · Reviewers · Delivery teams

BUSINESS CONSEQUENCE

Longer planning cycles · Rework · Unsupported scope · Lower delivery confidence

THE PROBLEM TO SOLVE

Convert fragmented evidence into consistent planning artifacts—without losing decisions, ownership, or source lineage.

Each manual pain point maps to a governed agent capability

The design automates translation work while keeping evidence, decisions, and release authority visible.

MANUAL FRICTION	GOVERNED CAPABILITY	EVIDENCE FLOW	RISK CONTROL
Files must be found and normalized	Drive Intake + Clarification Gate	Source files → normalized evidence	Hashing, required-file checks, clarification gate
Requirements are interpreted by hand	Requirement Extractor + Gap Analyzer	Citations → requirements → gaps	Citation IDs, confidence, contradiction and missing-data flags
PRDs and backlogs drift apart	Production PRD + Story Breakdown	Approved evidence → PRD → 3/7/11 hierarchy	Human approval, acceptance-criteria mapping, stable IDs
Release review is subjective	Final Validator + Story Sizing	Artifacts → reconciliation → sizing → export	Set equality, Agreement Gate, fail-closed export, non-blocking sizing

DESIGN PRINCIPLE

Automation accelerates the work; governance determines what may be delivered.

The charter defines a controlled product-documentation program

VISION

Transform fragmented stakeholder evidence into trusted, planning-ready product documentation with visible human control.

PROGRAM OBJECTIVES

01

Reduce translation effort

Automate evidence intake, extraction, gap analysis, and artifact generation.

02

Preserve decision lineage

Maintain source-to-requirement-to-story traceability across every stage.

03

Increase delivery confidence

Release only reconciled artifacts that satisfy deterministic and semantic controls.

MEASURABLE SUCCESS

100%

governed ID reconciliation

0

orphaned delivery records

11/11

stories with sizing guidance

3

validated planning artifacts

IN SCOPE

Evidence intake · requirement extraction · gap analysis · approval · PRD · stories · validation · sizing · export

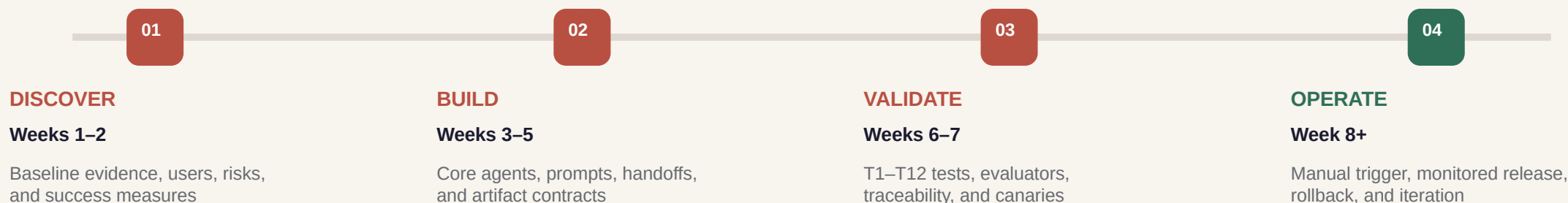
OUT OF SCOPE

Autonomous product decisions · automatic production triggers · engineering estimates as commitments · source-system replacement

Delivery is governed through ownership, stages, and rollout controls

The program moves from discovery to controlled operation without bypassing human approval or evidence validation.

PHASED DELIVERY PLAN



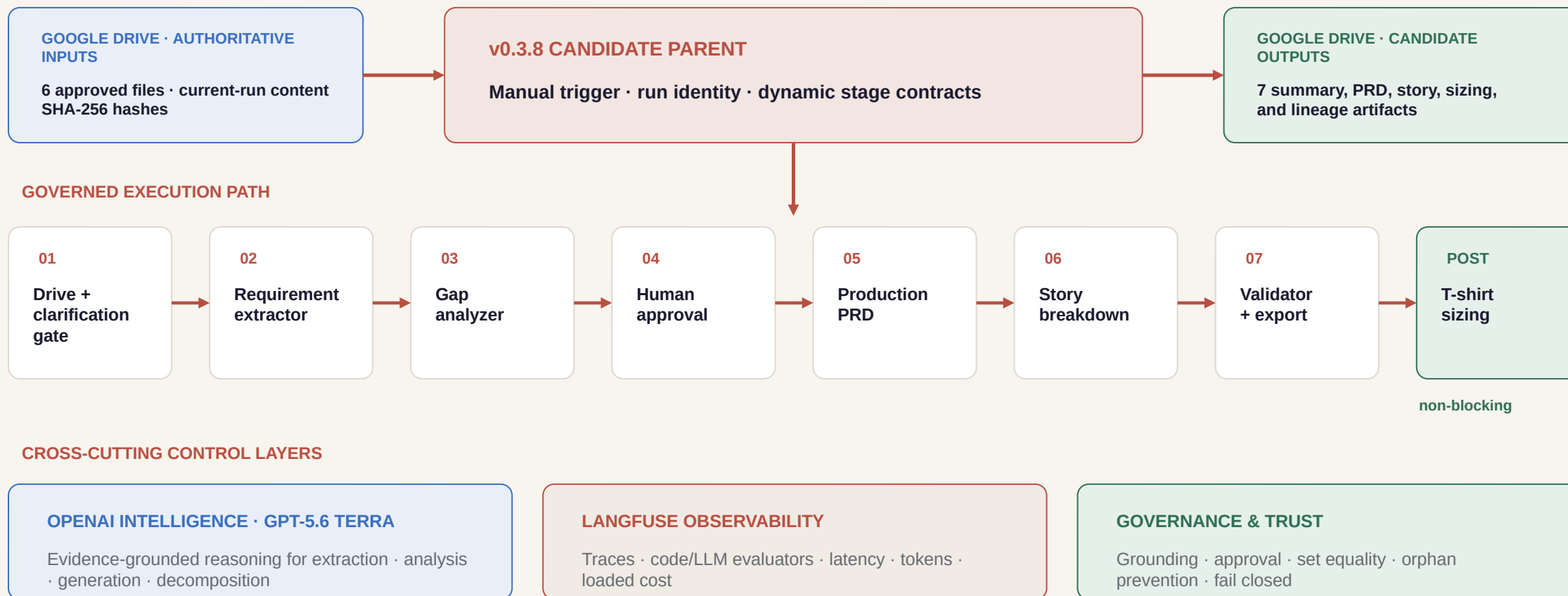
ACCOUNTABLE STAKEHOLDERS

Product owner	scope, approval, release decision
SME reviewers	evidence quality and clarification
Engineering team	artifact usability and feasibility
Workflow owner	operations, credentials, monitoring, rollback

PRINCIPAL RISKS → MITIGATIONS

Unsupported requirements	→	citation IDs + faithfulness / hallucination checks
Automation bias	→	mandatory human approval + visible uncertainty
Workflow regression	→	versioned JSONs + canary evidence + v0.3.7 rollback
Evaluator cost growth	→	deterministic-first controls + selective LLM evaluation

A sequential multi-agent architecture preserves evidence through delivery



One run identity connects every source, decision, artifact, evaluation, and export event.

Specialized workflows separate intelligence, judgment, and control

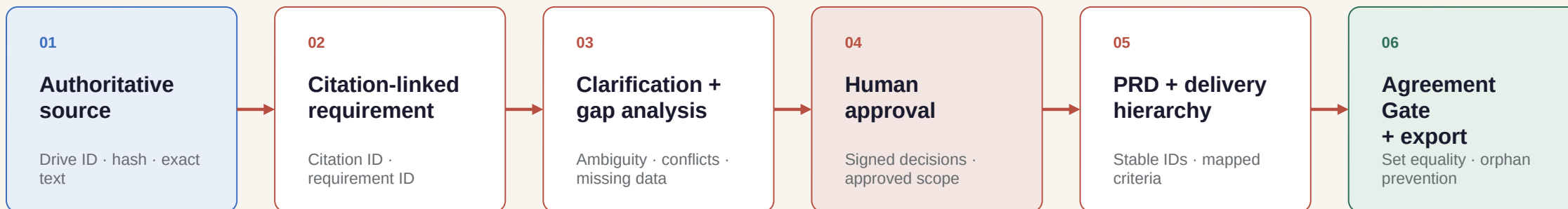
Each workflow receives a bounded contract and produces evidence for the next stage—rather than sharing one unrestricted prompt.

WORKFLOW	PRIMARY RESPONSIBILITY	INPUT → OUTPUT CONTRACT	EXECUTION RATIONALE
00 Candidate Parent	Create and govern one run	Source references → dynamic stage contracts	n8n deterministic orchestration; no content generation
01 Drive + Clarification	Establish evidence boundary	Files → normalized, hashed segments	Google Drive + n8n checks before interpretation
02 Requirement Extractor	Extract structured requirements	Segments → citation-linked items	GPT-5.6 Terra · grounded semantic extraction
03 Gap Analyzer	Expose gaps and conflicts	Requirements → gap and clarification register	GPT-5.6 Terra · conflict reasoning; uncertainty visible
04 Human Approval	Authorize scope and decisions	Evidence + gaps → signed dispositions	Human judgment is the production boundary
05 Production PRD	Generate corrected PRD	Approved evidence → ten-section Markdown PRD	GPT-5.6 Terra · grounded PRD generation
06 Story Breakdown	Create delivery hierarchy	PRD → 3 Epics / 7 Features / 11 Stories + AC	GPT-5.6 Terra · hierarchy and stable-ID reasoning
07 Validator + Export	Reconcile and release	Artifacts → authorized delivery files	Code controls + Agreement Gate + fail-closed export
POST T-shirt Sizing	Provide advisory effort	Stories → size, confidence, refinement guidance	GPT-5.6 Terra · advisory sizing; non-blocking

Grounding and human approval prevent unsupported requirements

Every generated element must carry evidence lineage, approved intent, and a reconciled delivery identity before export.

FAIL-CLOSED ASSURANCE PATH



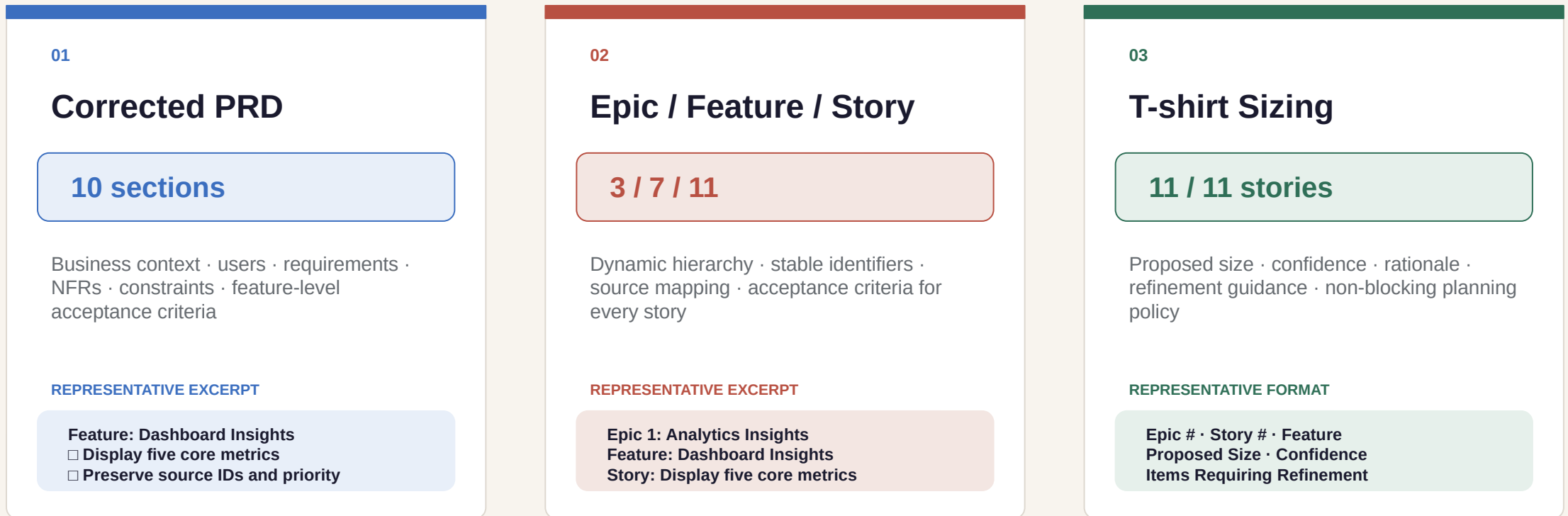
ACCEPTED v0.3.8 PROOF



No evidence → no requirement · No approval → no production · No reconciliation → no export

The pipeline produces three planning-ready artifacts

Each Markdown document is run-specific, human-readable, and backed by structured lineage and validation evidence.



All 12 prescribed baseline scenarios are tested

Independent ground truth verifies extraction behavior, document generation, and delivery decomposition.

12 / 12

approved baseline cases

The baseline tests what the system does when inputs are clear, ambiguous, conflicting, incomplete, highly specific, or empty—before validating the two generated planning documents.



T1–T10

Requirement extraction

Detailed · vague · contradictory · incomplete · persona · technical · dependency · empty inputs

10 / 10 controls passed



T11

PRD generation

Approved evidence → grounded ten-section PRD

PASS



T12

Story breakdown

Approved PRD → Epic → Feature → Story → criteria

PASS

THE BASELINE PROVES

Exact values preserved · Ambiguity flagged without invention · Acceptance criteria mapped · Evidence captured

Deterministic and LLM evaluators measure different failure modes

Release confidence comes from combining exact contract checks with evidence-grounded semantic judgment.

LENS 01 · EXACT CONTRACT

Deterministic code controls

Answers: Did the workflow obey the required structure and evidence contract?

-
- | | |
|-----------------------------------|--------------------------|
| ✓ Schema + required fields | correct shape |
| ✓ Set equality + counts | nothing missing or extra |
| ✓ Citation + ID validity | valid mappings |
| ✓ Orphans + export rules | fail closed |

LENS 02 · SEMANTIC QUALITY

GPT-5.6 Terra semantic evaluators

Answers: Is the content supported, sensible, and free from material invention across the full evidence context?

-
- | | | |
|-------------------------|-----------------------------------|-----------------|
| ● Faithfulness | Supported by approved evidence | maximize |
| ● Hallucination | Unsupported material claims | ≤ 0.15 |
| ● Reasonableness | Planning logic and sizing quality | maximize |

DECISION LAYER

Agreement Gate combines both lenses: release only when required code controls pass and semantic scores remain within policy.

LANGFUSE EVIDENCE

OBSERVABILITY

Every accepted run retains stage traces, evaluator receipts, latency, token usage, and fully loaded cost evidence.

The accepted v0.3.8 run passed end to end

One governed run produced the complete document package, passed evaluation, and received release authorization.



RELEASE AUTHORIZED

Agreement Gate passed · final validator passed · 7 artifacts delivered · sizing completed 11/11

v0.3.8

ACCEPTED RUN IDENTITY

One identity connects every stage

The parent execution and run ID connect source files, stage handoffs, evaluator receipts, output documents, and export evidence.

Parent execution	11901
Run ID	RUN-S2-11902-16e7090e
Duration	2m 33.201s

STAGE-LEVEL EVALUATION

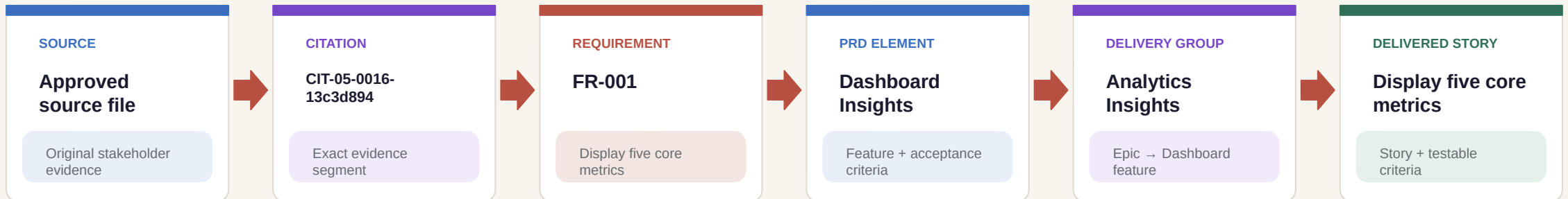
Generated artifacts remained grounded

STAGE		CODE	FAITHFUL	HALLUC.	
	Production PRD	ten-section structure	PASS	0.99	0.05
	Story Breakdown	11 stories reconciled	1.00	0.98	0.02
	Story Sizing	reasonableness 1.00	1.00	1.00	0.00

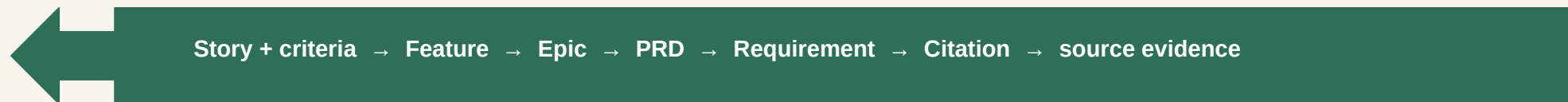
Every delivered story can be traced back to source evidence

Stable identifiers preserve the chain in both directions—from approved source text to acceptance criteria and back.

FORWARD TRACE · Why does this story exist?



REVERSE TRACE · Which evidence supports this delivered item?

**145**

citations indexed

145 / 145

terminal dispositions

0

orphan citations

0

orphan PRD elements

0

orphan delivery items

Fully loaded cost establishes an operational baseline

The accepted baseline deliberately uses GPT-5.6 Terra where grounded reasoning and evaluation quality are release-critical.

\$1.474409

fully loaded cost per accepted run

545,467

observed tokens

1 run / week

≈ \$6.38 per user / month

COST COMPOSITION

Semantic evaluation drives most of the loaded cost

COMPONENT	TOKENS	COST
Pipeline generation Core agent generation and transformation	59,922	\$0.228024
LLM evaluator calls Faithfulness, hallucination, reasonableness	485,545	\$1.246385

OPERATING INTERPRETATION

84.5%

of cost is semantic evaluation



15.5% generation

84.5% evaluation

Quality-first baseline: retain GPT-5.6 Terra for high-risk reasoning; test lighter models on lower-risk tasks in v0.3.9.

COST BOUNDARY

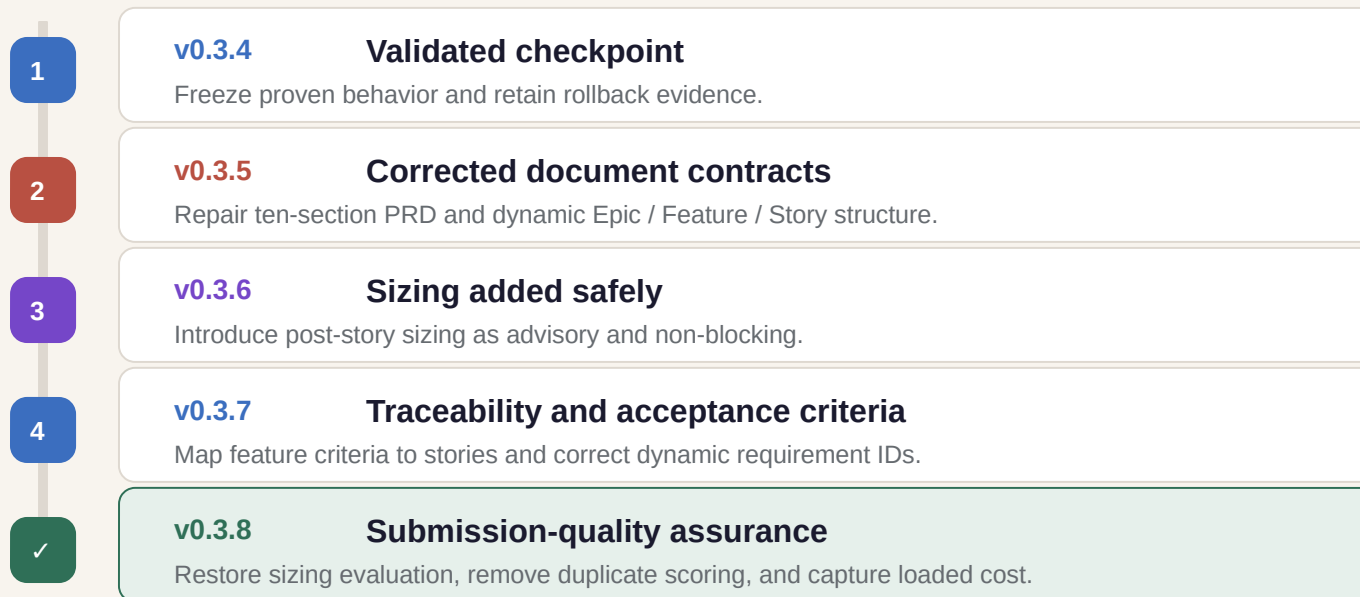
Observed model and evaluator usage for the accepted run; platform subscriptions and optional enterprise services are separate.

Iteration converted observed failures into production controls

Each release preserved a working checkpoint, isolated one correction, and added evidence before promotion.

CONTROLLED LEARNING PATH

Five releases turned failure evidence into safeguards



WHAT THE EVALUATION TAUGHT

AI output needs governed boundaries

Unsupported detail

Ground every claim and preserve visible uncertainty.

Silent mapping loss

Reconcile citation, PRD, and delivery sets both ways.

Evaluator cost

Use code for exact checks; reserve LLM judgment for semantics.

v0.3.9 NEXT ITERATION

Test lighter models on lower-risk tasks; retain GPT-5.6 Terra where reasoning remains release-critical.

The submission package makes every claim inspectable

A single evidence index connects the written answers, implementation, accepted run, product outputs, and demo.

REPRODUCIBLE SUBMISSION PACKAGE

01

Implementation

v0.3.8 workflow JSONs · inventory · configuration notes

02

Written responses

Q1–Q4 PDFs / Markdown · README · rubric cross-reference

03

Baseline testing

T1–T12 matrix · inputs · expected checks · result evidence

04

Product outputs

Corrected PRD · Epic / Feature / Story · sizing report

05

Runtime evidence

n8n IDs · Langfuse traces · scores · tokens · loaded cost

06

Presentation + demo

PPTX / PDF · complete execution and evidence walkthrough

WHAT A REVIEWER CAN VERIFY

One path from rubric claim to evidence



Rebuild the topology

Open the nine v0.3.8 workflow definitions and call graph.



Inspect the accepted run

Resolve execution 11901 and RUN-S2-11902-16e7090e.



Trace evidence to delivery

Follow source → citation → requirement → PRD → story.



Validate quality and cost

Check evaluator results, 545,467 tokens, and \$1.474409.

READY

Evidence preserved locally and in platform records

PRD Orchestrator is ready for governed use

The accepted v0.3.8 candidate converts fragmented source evidence into planning-ready artifacts without losing control of provenance, quality, or release.

RECOMMENDATION

Accept v0.3.8 as the submission candidate and preserve v0.3.7 as the protected rollback checkpoint.

01

Evidence remains connected

145 indexed citations resolve to terminal dispositions with zero orphan citations, PRD elements, or delivery items.

02

Release remains controlled

Human approval, deterministic controls, semantic evaluation, and the Agreement Gate authorize export only after reconciliation.

03

Outputs are planning-ready

One accepted run delivers the corrected PRD, Epic / Feature / Story breakdown, and advisory 11/11 sizing evidence.

ACCEPTED RUN

11901 · Agreement Gate authorized · 7 artifacts · sizing 11/11



SUPPORTING EVIDENCE

Appendix

Detailed evidence behind the architecture, controls, evaluation, traceability, cost, and release decisions.

WORKFLOW INVENTORY

Names, IDs, call graph, and versioned runtime path

EVALUATION & COST

Test evidence, evaluator results, tokens, and loaded cost

TRACEABILITY & RELEASE

Citation lineage, artifacts, version history, and rollback

Every scored requirement maps to inspectable evidence

The main deck communicates the argument; the written package and platform records provide the verification path.

ASSIGNMENT	PTS	DECK	PRIMARY EVIDENCE
Q1 — Ideation	15	3–4	Pain-point analysis; manual step → agent capability → input / output → risk and control
Q2 — Program Charter	15	5–6	Vision, objectives, scope, success criteria, ownership, timeline, risks, and rollout plan
Q3 — Build	45	7–15	Architecture; agents; controls; outputs; T1–T12; code + LLM evaluation; accepted-run results; traceability; loaded cost
Q4 — Reflection	5	16	Observed failures, production controls, AI risks, lessons learned, and v0.3.9 improvement plan
TOTAL COVERAGE			
80 / 80 points mapped			
Written evidence + deck + runtime receipts			

Q3 · 45-POINT DETAIL

Build evidence coverage

10	Architecture, orchestration, tool rationale	7–8
12	Core capabilities end to end	9–10,13
8	Extended capability	9, 14
5	Observability connected	12–13
5	Baseline dataset tested	11
5	Cost and evaluation strategy	12,15

45 / 45 points represented

Nine versioned workflows form the v0.3.8 candidate set

The accepted parent uses a minimal-change hybrid call graph while the complete candidate copy remains exportable and auditable.

#	v0.3.8 CANDIDATE WORKFLOW	LIVE n8n ID	DISPOSITION
00	PRD Genie — Main Orchestrator — v0.3.8 Candidate	YCgHHBa8xUvSOYGI	Accepted parent
01	PRD Genie — Drive Intake + Clarification Gate — v0.3.8 Candidate	CQEgz6G2sPHhehjW	Snapshot copy
02	PRD Genie — Requirement Extractor — v0.3.8 Candidate	YcNGjVHdyAJgZeod	Snapshot copy
03	PRD Genie — Gap Analyzer — v0.3.8 Candidate	6F4ANeRIDJVKA28C	Snapshot copy
04	PRD Genie — Human Approval — v0.3.8 Candidate	uMSo2zvHLGWcM2vo	Snapshot copy
05	PRD Genie — Production PRD — v0.3.8 Candidate	ocoEGqFDyzzFYf3U	Invoked
06	PRD Genie — Story Breakdown — v0.3.8 Candidate	kswPN0mT0u7rp4vq	Invoked
07	PRD Genie — Final Validator + Export — v0.3.8 Candidate	gU24pjEsig7u60S4	Snapshot copy
08	PRD Genie — Post-Stage-7 Story Sizing — v0.3.8 Candidate	cFuv8QCLpLhtX6A6	Invoked

AUTHORITATIVE RUNTIME PATH

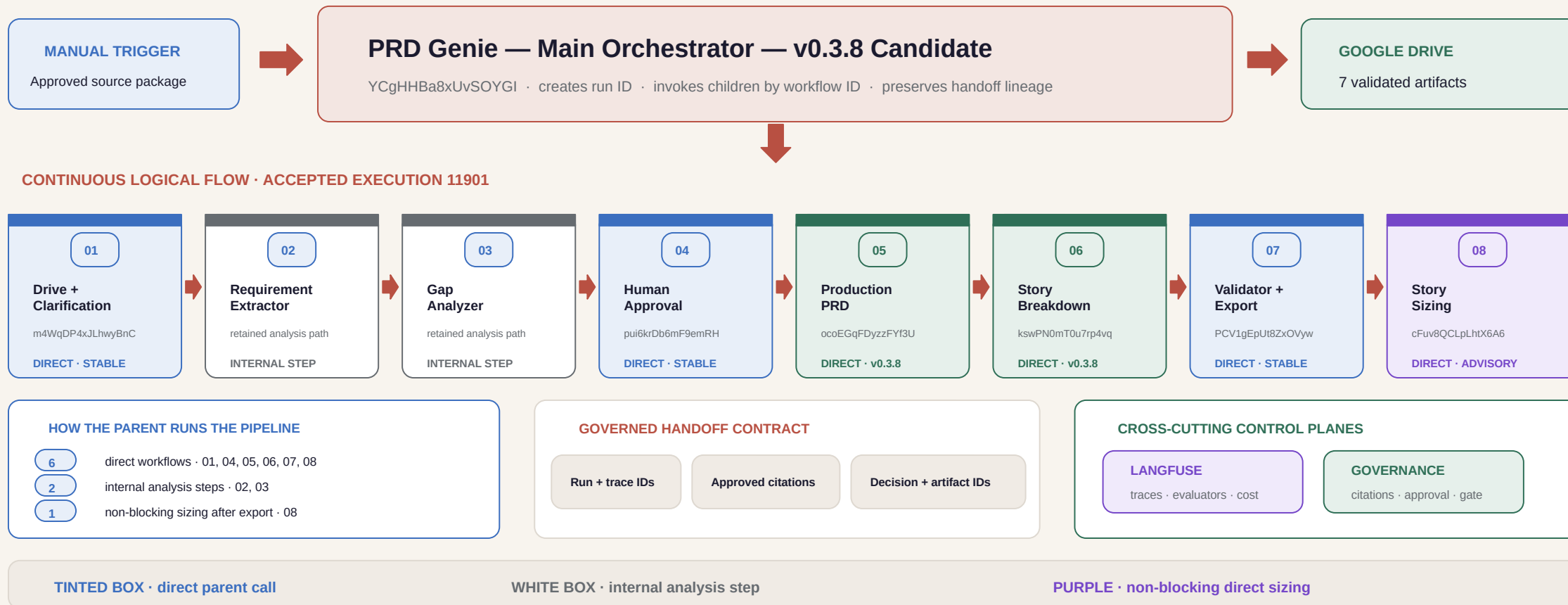
Accepted hybrid call graph

Stable controls remain in place; corrected generation and sizing stages use v0.3.8 IDs.

01	Drive + Clarification m4WqDP4xJLhwyBnC	STABLE
04	Human Approval pui6krDb6mF9emRH	STABLE
05	Production PRD ocoEGqFDyzzFYf3U	v0.3.8
06	Story Breakdown kswPN0mT0u7rp4vq	v0.3.8
07	Validator + Export PCV1gEpUt8ZxOVyw	STABLE
08	Story Sizing cFuv8QCLpLhtX6A6	v0.3.8
Execution 11901 · RUN-S2-11902-16e7090e		

One governed sequence connects all eight logical stages

Six stages are direct child calls; Requirement Extraction and Gap Analysis remain retained logical contracts within the analysis path.



Prompts constrain generation; independent gates control release

The model may propose content, but it cannot authorize unsupported requirements or bypass evidence, approval, and reconciliation controls.

AGENT PROMPT CONTRACT

- 01

Use approved evidence only
No requirement, date, priority, dependency, or product fact may be invented.
- 02

Attach exact citations
Every factual output resolves to a source ID, evidence segment, and requirement ID.
- 03

Preserve uncertainty
Ambiguity, contradiction, and missing information remain visible until resolved.
- 04

Honor structured contracts
Return schema-valid JSON and synchronized Markdown with stable identifiers.
- 05

Respect authority boundaries
Extraction, analysis, approval, generation, validation, and sizing retain distinct roles.



INDEPENDENT ENFORCEMENT

- INPUT INTEGRITY**
Source hashes, exact lines, unique IDs
- DETERMINISTIC VALIDATION**
Schema, coverage, set equality, zero orphans
- HUMAN APPROVAL**
Signed decisions and approved scope lineage
- SEMANTIC EVALUATION**
Faithfulness, hallucination, reasonableness
- AGREEMENT GATE**
Release only when code and LLM controls agree

ACCEPTED RELEASE CONDITION	100% groundedness	0 unsupported claims	Fail closed on mismatch
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Twelve prescribed cases test extraction through delivery

Human-reviewed ground truth covers precise, vague, contradictory, incomplete, persona, technical, dependency, PRD, and story-generation behavior.

T1-T6 · REQUIREMENT EXTRACTION

T1	Precise mixed requirements Complete · 1 FR, 1 NFR, stakeholder, deadline	EXECUTION RECEIPT LF f14cab...0002	PASS
T2	Vague comparative request Partial · broad FR + 4 clarification records	EXECUTION RECEIPT LF 114917...4b95	PASS
T3	Contradictory refresh goals Partial · FR, NFR, contradiction preserved	EXECUTION RECEIPT LF 1d107c...7d9d	PASS
T4	Export acceptance criteria Complete · 1 FR + 2 testable criteria	EXECUTION RECEIPT LF 119cf3...0901	PASS
T5	Fragmentary uncertain notes Partial · no invention + 3 clarifications	EXECUTION RECEIPT LF e6fd72...38e1	PASS
T6	Conflicting technical direction Partial · 2 constraints, deadline, contradiction	EXECUTION RECEIPT LF 4df943...f4d7	PASS

T7-T12 · TECHNICAL, PRD & STORY CONTROLS

T7	Exact technical metrics Complete · 3 exact-value NFRs preserved	EXECUTION RECEIPT LF f1f4b6...f56f	PASS
T8	Personas and permissions Complete · 3 personas + 3 matched FRs	EXECUTION RECEIPT LF 29a078...ab6e	PASS
T9	Empty meeting notes No requirements · generation correctly refused	EXECUTION RECEIPT LF 3c54e4...205c	PASS
T10	Dependency with unknown ETA Complete · FR, dependency, risk; ETA unknown	EXECUTION RECEIPT LF 0c99f0...1bb5	PASS
T11	Ten-section PRD generation 10/10 sections · approved IDs only	EXECUTION RECEIPT n8n 8621 · LF 05e9...48ff	PASS
T12	Epic-to-story decomposition 1 epic, 1 feature, 1 story, 2 criteria	EXECUTION RECEIPT n8n 8788 · LF 8e20...159b	PASS

12 / 12

required cases passed

100% groundedness

0 unsupported claims

12 Langfuse traces preserved · T1-T10 successful run metadata · T11 n8n 8621 · T12 n8n 8788

Every extraction test preserves the supplied facts and uncertainty

Exact source strings are paired with the actual structured fields emitted by the validated Requirement Extractor release.

EXACT BASELINE INPUT	ACTUAL PRESERVED OUTPUT	EXECUTION RECEIPT
T1 "The user should be able to filter reports by date range, category, and status. Results must load in under 2 seconds. PM: Sarah. Deadline: Q3."	complete · FR-001 report filtering · NFR-001 <2 sec · STK-001 Sarah · DDL-001 Q3	RUN-T1-1785799870489 LF f14cab...0002 PASS
T2 "We need better reporting. Something like what Competitor X has."	partial · FR-001 better reporting · MISS-001 users · MISS-002 success definition · MISS-003 format/delivery · MISS-004 competitor scope	successful metadata LF 114917...4b95 PASS
T3 "The dashboard should auto-refresh every 5 seconds. [Later in same meeting] Performance is critical, minimize API calls."	partial · FR-001 5-sec refresh · NFR-001 minimize API calls · CTR-001 conflict preserved unresolved	successful metadata LF 1d107c...7d9d PASS
T4 "Users need to export reports as PDF and CSV. PDF must include company logo. CSV must preserve formulas."	complete · FR-001 PDF/CSV export · AC-001 PDF logo · AC-002 CSV formulas preserved	successful metadata LF 119cf3...0901 PASS
T5 "Discussed dashboard... John mentioned something about real-time... budget TBD..."	partial · no unsupported requirement · MISS-001 dashboard scope · MISS-002 real-time meaning · MISS-003 budget	successful metadata LF e6fd72...38e1 PASS
T6 "Engineering wants microservices. Design wants single-page app. PM wants it shipped by March."	partial · CON-001 microservices · CON-002 SPA · DDL-001 March · CTR-001 architecture conflict unresolved	successful metadata LF 4df943...f4d7 PASS
Method: human-reviewed expected output → preserved actual JSON/Markdown ↔ successful n8n/Langfuse receipt		

Technical constraints flow into grounded PRD and story artifacts

The remaining cases verify exact metrics, personas, refusal behavior, dependencies, PRD generation, and story decomposition.

EXACT BASELINE INPUT	ACTUAL PRESERVED OUTPUT	EXECUTION RECEIPT
T7 "API must support 10,000 concurrent users. Response time < 200ms at p95. Must integrate with Salesforce REST API v52."	complete · NFR-001 10,000 users · NFR-002 <200 ms p95 · NFR-003 Salesforce REST API v52	successful metadata LF f1f4b6...f56f PASS
T8 "Admins need bulk user management. End users need a simplified view. Auditors need read-only access with full history."	complete · PER-001/FR-001 Admins · PER-002/FR-002 End users · PER-003/FR-003 Auditors	successful metadata LF 29a078...ab6e PASS
T9 "Meeting happened. Notes: none."	no_requirements · generation refused · MISS-001 requests an authoritative source with product details	successful metadata LF 3c54e4...205c PASS
T10 "SSO login requires the new auth service which is being built by Team Alpha. ETA unknown."	complete · FR-001 SSO/auth service · DEP-001 Team Alpha, ETA unknown · RSK-001 unknown ETA	successful metadata LF 0c99f0...1bb5 PASS
T11 Approved T1 package: FR-001 report filtering; NFR-001 results <2 sec; STK-001 Sarah; DDL-001 Q3; signed Human Approval.	Ten-section PRD · FR-001 + NFR-001 · AC-001 + AC-002 · Q3 timeline · TBD fields remain explicit · 100% grounded	n8n 8621 LF 05e9aa...48ff PASS
T12 Approved T11 PRD: FR-001, NFR-001, AC-001, AC-002; source run RUN-T1-GROUND-TRUTH.	EPIC-001 Report Filtering → FEAT-001 Filter Reports → US-001 → AC-001 + AC-002 · 2 unresolved questions · 0 unsupported claims	n8n 8788 LF 8e2078...159b PASS
Method: human-reviewed expected output → preserved actual JSON/Markdown ↔ successful n8n/Langfuse receipt		

Code controls and semantic judges cover different failure modes

Accepted v0.3.8 run 11901 · RUN-S2-11902-16e7090e · one current-trace Story score set authorized release.

STAGE	CODE EVALUATOR / RESULT	FAITHFUL	HALLUC.	OUTCOME
Requirement Extractor	prd_genie_s2_code_controls_v1 · Pass	—	—	PASS
Gap Analyzer	prd_genie_gap_analyzer_code_controls_v1 · Pass	0.90	0.01	PASS
Human Approval	ha_code_controls_v2 · Pass	0.99	0.00	PASS
Production PRD	prd_genie_production_prd_code_controls_v1 · Pass	0.99	0.05	PASS
Story Breakdown	prd_genie_story_breakdown_code_controls_v1 · Pass	0.98	0.02	GATE PASS
Sizing · 11 stories	sizing_lf_code_evaluation_pass · 11/11	1.00 avg	0.00 avg	REASON 1.00

EVALUATION DESIGN

One control stack, two kinds of proof

CODE · deterministic

Schema · counts · exact values · mappings · required sections · release contracts

LLM · semantic

Generic: llm_faithfulness + llm_hallucination
Stage-specific: story_* + sizing faithfulness, hallucination, reasonableness

Policy: faithfulness ≥ 0.90 · hallucination ≤ 0.15 · code = 1

ACCEPTED-RUN EVIDENCE RECEIPTS

Story trace 35579c15ebbeb1bf83406a8274600333 Sizing trace 0a29e318a6784852cc64b997229b00c2

Langfuse project cms9sudes0u9rad0h4fvs3r6n · Evaluations

Agreement Gate: complete score set · no duplicates · current-run correlation · release authorized

Semantic evaluation accounts for 84.5% of the accepted-run cost

Actual Langfuse usage for v0.3.8 execution 11901 · generation plus every completed current-run LLM evaluator.

\$1.474409

fully loaded cost

545,467

total tokens

1 run / week

governed operating assumption

~\$6.38 / month

4.33 fully evaluated runs

COST SHARE



COST COMPONENT	CALLS	INPUT	OUTPUT	TOTAL TOKENS	ACTUAL COST
Pipeline generation subtotal	2	49,104	10,818	59,922	\$0.228024
Generic faithfulness + hallucination	6	310,227	1,690	311,917	\$0.795840
Story Breakdown faithfulness + hallucination	2	144,903	531	145,434	\$0.368627
Sizing: faithfulness + hallucination + reasonableness	33	25,641	2,553	28,194	\$0.081918
LLM evaluator subtotal	41	480,771	4,774	485,545	\$1.246385
FULLY LOADED TOTAL	43	529,875	15,592	545,467	\$1.474409

CONTROL STRATEGY

Code checks first · no duplicate semantic scoring · stage-specific judges only where needed · full evaluation retained for candidate acceptance

Every active delivery row resolves to evidence in both directions

Accepted v0.3.8 run 11901 · source → citation → PRD → Feature AC → Epic → Feature → Story → Story AC · and back.

GITHUB EVIDENCE Complete 145-citation ledger → github.com/vppuri-vjra/prd-genie/tree/main/releases/v0.3.8/evidence

CITATION ID(S)	PRD ID	FEATURE AC	EPIC	FEATURE	STORY	STORY AC	STATUS
CIT-05-0016-13c3d894	FR-001	FAC-001	EPIC-001	FEAT-001	US-001	SBAC-001	ACTIVE
CIT-05-0017-f0ae7fbb · CIT-01-0012-f394e26d	FR-002	FAC-002	EPIC-001	FEAT-002	US-002	SBAC-002	ACTIVE
CIT-04-0016-40b68902 · CIT-04-0013-23ba34cb	FR-006 · CON-002	FAC-003	EPIC-001	FEAT-003	US-003	SBAC-003	ACTIVE
CIT-04-0016-40b68902 · CIT-04-0013-23ba34cb	FR-006 · CON-002	FAC-004	EPIC-001	FEAT-003	US-004	SBAC-004	ACTIVE
CIT-04-0016-40b68902	FR-006	FAC-005	EPIC-001	FEAT-003	US-005	SBAC-005	ACTIVE
CIT-04-0016-40b68902	FR-006	FAC-006	EPIC-001	FEAT-003	US-006	SBAC-006	ACTIVE
CIT-05-0019-192d5768	FR-003	FAC-007	EPIC-002	FEAT-004	US-007	SBAC-007	ACTIVE
CIT-05-0019-192d5768	FR-003	FAC-008	EPIC-002	FEAT-004	US-008	SBAC-008	ACTIVE
CIT-02-0009-2d08dd05	FR-007	FAC-009	EPIC-002	FEAT-005	US-009	SBAC-009	ACTIVE
CIT-05-0020-1d09d883 · CIT-01-0074-640ddf26	FR-004 · AC-001	FAC-010	EPIC-003	FEAT-006	US-010	SBAC-010	ACTIVE
CIT-04-0015-1afd65a7	FR-005	FAC-011	EPIC-003	FEAT-007	US-011	SBAC-011	ACTIVE

145 / 145

citation dispositions

26

citations used in PRD

20

PRD requirement IDs

11 / 11

Feature AC → Story AC

31 / 31

approved-item coverage

0

orphans or unsupported claims

Seven accepted files make the v0.3.8 delivery package inspectable

First accepted candidate · execution 11901 · RUN-S2-11902-16e7090e · Agreement Gate: Release Authorized

GITHUB EVIDENCE PACKAGE → github.com/vppuri-vjra/prd-genie/tree/main/releases/v0.3.8/evidence

01 Validated Full PRD Review

Corrected ten-section PRD with feature-level acceptance criteria and evidence-backed requirements.

10 sections · 20 requirement IDs · 35 PRD elements

Drive ID 1NZZJgu1UBvyB91jr8wQ72FO8RCP0wING

02 Validated Epic / Story Review

Dynamic Epic–Feature–Story decomposition with mapped story acceptance criteria.

3 Epics · 7 Features · 11 Stories · 11 Story AC

Drive ID 1avNdNgVi1cDIBvBC-6YNupgsYgPvEaJT

03 Proposed T-Shirt Sizing Review

Non-blocking planning guidance with proposed size, confidence, and refinement needs.

11 / 11 stories sized · advisory after export

Drive ID 1a3jiUoNy2bQ0mvr6aZpFDIUPTz9ONupb

MACHINE AND AUDIT ARTIFACTS

Run summary

JSON

1KUzolzV9l0Ddz86tgLJeryXU5VyWVIJS

Final PRD

MARKDOWN

10Rc9rtw-zYemiswLM0aPnoLIHmDtSRVI

Story breakdown

JSON

1p0C48qQmkN88DaLQ-urdEsOZI2nUaKqr

Traceability and lineage

JSON

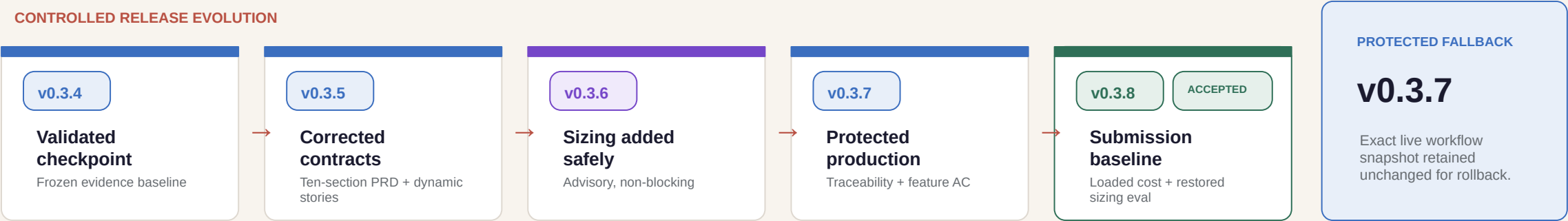
1F92GAjQcMCKG-T-7EwBDsIDRTDWwnG3N

One run identity connects every source package, approval decision, generated document, evaluator result, and exported file.

v0.3.8 advances the system without sacrificing a proven rollback path

Each promotion preserves a versioned workflow set, accepted-run evidence, and an unchanged fallback before the next correction is introduced.

CONTROLLED RELEASE EVOLUTION



ROLLBACK RECEIPTS

The prior production state is recoverable—not reconstructed

Exact workflow snapshot	c5ae311	Nine v0.3.7 live-production JSON exports
Merged checkpoint	57618f6 · PR #16	Versioned evidence preserved on main
Protected controls	Manual trigger · Agreement Gate	Stable intake, approval, validator, and export

v0.3.8 PROMOTION RECEIPTS

077f77c

accepted submission package merged to main

Execution 11901 · **RUN-S2-11902-16e7090e**

7 outputs · 11/11 sizing · Release Authorized

Promotion is reversible because code, workflow IDs, run evidence, and the prior production checkpoint are preserved together.